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Anti-fatigue shoe apparatus

Abstract

An anti-fatigue shoe apparatus is presented that is designed to reduce foot fatigue and discomfort. The anti-fatigue shoe apparatus makes use of novel combinations of insole, midsole, outsole, and comfort material configurations and designs to provide comfort, support, and flexibility within work footwear. The anti-fatigue shoe apparatus includes a raised arch support piece that cradles the foot's natural arch, reducing torque and improving stability. Toe and heel comfort zones made of cushioning materials provide targeted support and pressure relief. In some embodiments an injection system for providing a liquid comfort material injected into the toe and heel zones through canals in the outsole are presented for customized cushioning. The invention comes in various material combinations.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to and the benefit of U.S. Provisional Patent Application No. 63/449,661, filed Mar. 3, 2023, which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

(1) The present invention pertains to an anti-fatigue shoe apparatus for long term usage and to improving the comfort of a worker who is required to stand for multiple long hours per day. The present invention further pertains to an apparatus that combines comfort and strength to a worker's foot. The apparatus is designed without the use of hard and overly sturdy materials that create extremely limited flexibility of shoes and work boots. The present invention utilizes novel combinations of insole, midsole, and outsole designs that provide comfort and flexibility within shoes for workers, so that comfort and support are achieved when long term static pressure on or long hours walking in work footwear are required.

DESCRIPTION OF THE RELATED ART

(2) Most work footwear soles are made from hard and sturdy materials due to the need for protection and durability, such materials provide very little flexibility. Work footwear that provides increased flexibility and adequate support to workers who are required to stand for multiple hours per day and/or walk several miles per day, over long periods of time, are important for the long-term health of workers. Having access to footwear that provides workers' feet with the ability to rest in the natural position while in a standing position while also providing enough support for adequate movement is increasingly necessary.

(3) Work footwear has been available in the market for quite some time. Existing footwear is available in a multitude of various shapes, sizes, harnesses, usage purposes, and stiffnesses, all of which have the basic utility of providing comfort and support to users' feet. There have also been many variations of footwear designed for workers with the same basic utility.

(4) The vital nature of providing proper support and comfort to workers' feet will continue to remain significant. These basic requirements are critical for all humans to preserve health and reduce the risk of various injuries. Perhaps the greatest problem that exists with work footwear is that currently available products on the market provide little flexibility and/or are designed solely for fashion. Additionally, currently available work footwear is not specifically designed for workers that are required to stand for long periods of the day nor for workers that are required to walk multiple miles per day. It is the goal of the present invention to use novel combinations of insole, midsole, outsole, and comfort material designs to provide comfort, support, and flexibility within work footwear, such that comfort and support are achieved when long term static pressure within or long hours of walking in work footwear are required.

(5) U.S. Pat. No. 10,772,381 B2 discloses a shoe insole that includes a semi-rigid shell that includes a notch located on the outward side of the wearer's foot to accommodate pronation of the foot during use for providing a wearer with extended comfort. Additionally, the semi-rigid shell includes an extended side portion located opposite of the notch side on an inner side of the wearer's foot. The semi-rigid shell is shorter in length than the length of a shoe sole and slightly shorter in width and may be combined with an insole comfort cover to provide cushion between the wearer's foot and the semi-rigid shell.

(6) U.S. Pat. No. 6,516,541 B2 discloses a toddler's and children's shoe meeting a parent's need for

fashionable decoration of the toddler, while simultaneously allowing that child freedom and comfort of pedal movement, while avoiding repetitive stress injury to the foot. This reference discloses a shoe with a flexible outsole, an insole, an upper formed from a flat Thermo Plastic Rubber blank, and a toe cap fabricated in the blank via a teacup crease special-use sewing machine. The blank or preform subsequently affixed to a last and joined by a second special purpose sewing machine, or disc feed overseaming machine, to a non-woven fabric midsole or insole which, substantially completes the upper without use of an insole board. The outsole of the shoe is unitary in construction and equipped with a unique pattern of intersecting grooves, as well as, an external bridge or instep support in lieu of an inner steel shank.

(7) U.S. Pat. No. 4,930,232 A discloses a multilayer laminate for use as a sole of a fashion or dress shoe, wherein the shoe construction includes a combination of rigid support materials and polymeric foam materials of varying Shore hardness values permanently attached atop a conventional outer sole. The specific structural configuration of the material enhances the durability of the shoe and improves the comfort it provides for its wearer when engaged in walking, strolling, or other foot-utilizing activity.

(8) U.S. Pat. No. 7,464,490 B2 discloses a multilayered sole having an insole, a midsole, and an outsole, where the insole extends from a toe area to a heel area that has a rigid member and a flexible member for enhancing flexibility. The midsole extends from the toe area to the heel area and has a cushioning material and a structural material for enhancing comfort.

(9) U.S. Pat. No. 9,565,895 B2 discloses an article of footwear that has a molded outsole which contains a resilient midsole located in the cavity of the outsole, the midsole has sections of different hardness in the heel area and the forefoot area, and a footbed cover which has a stretch material layer affixed to the outsole and provides a stretch zone. The stretch zone and the footbed cover work in conjunction with the midsole to absorb shock upon heel strike or other compression of the midsole to increase comfort.

(10) U.S. Patent Application Publication 2012/0047768 A1 discloses a silicone gel insole/midsole for numerous types of footwear that is layered and bonded between the upper of a shoe and an extra-depth outsole. The silicone insole/midsole covers the entire top surface of the relatively rigid outsole of the footwear. The insole/midsole uses silicone gel material in functional relation with an article of footwear in order to enhance fit, support, comfort, cushioning and to mimic the energy dissipation properties of human soft tissue and structure.

(11) WIPO Patent Publication No. WO 2018/192262 A1 discloses an insole or a sole having a recess in the shape of a figure-of-eight and a soft cushioning layer covering the recess in contact with the soft cushioning located on the top surface of an insole body. The figure-of-eight-shaped recess includes a rear recess, a front recess corresponding to the position of a metatarsal bone, and a middle recess communicating with the rear recess and the front recess. The front end of the front recess inclines to the inner side of the insole body and the bottom surface of the front recess rises from the edge of the recess to the center of the recess to form a raised arc surface.

(12) All aforementioned patents and publications are incorporated herein by reference.

(13) While these devices may be suitable for the particular purpose employed, they would not be as suitable, or suitable at all, for the purposes of the present invention as disclosed hereafter.

(14) While the prior art discloses various devices and apparatuses for providing comfort to users of footwear, the present invention implements the use of novel combinations of insole, midsole, outsole, and comfort material configurations and designs to provide comfort, support, and flexibility within work footwear. This is advantageous because these novel configurations and designs can accommodate the comfort and support needs of workers that are required to stand for long periods of the workday, which results in significant static pressure on insoles, midsoles, and outsoles. Additionally advantageous aspects of the present invention also accommodate the comfort and support needs of workers that are required to walk several miles per day, which requires additional flexibility. Thus, the present invention ensures that workers with these needs do not

experience discomfort while also allowing for workers' feet to rest in the natural position by conforming with the natural shape of the workers' feet.

SUMMARY OF THE INVENTION

(15) It is one prospect of the present invention to provide a novel apparatus that combines comfort and strength to a worker's foot, the apparatus is designed for long-term usage by users who spend long hours in a standing position without the use of hard and overly sturdy materials that create extremely limited flexibility of shoes and work boots, the present invention utilizes a novel combination of insole, midsole, and outsole designs to provide comfort to users so that comfort and support are achieved when static pressure in the users' shoes and work boots are required.

(16) The following presents a simplified summary of the present disclosure in a simplified form as a prelude to the more detailed description that is presented herein.

(17) Therefore, in accordance with a preferred embodiment of the present invention, there is provided an anti-fatigue shoe apparatus. The anti-fatigue shoe apparatus has an outsole that has a top outsole surface that is opposite a bottom outsole surface. The bottom outsole surface is adapted to grip the ground. The anti-fatigue shoe apparatus has a midsole that is adapted to reduce torque. The midsole has a top midsole surface that is opposite a bottom midsole surface. In accordance with the present invention, a midsole arch cavity is centrally formed within the bottom midsole surface of the midsole and is located between a midsole toe section the is oriented laterally opposite of a midsole heel section. A raised arch shaped support piece is oriented within and adhered to the bottom midsole surface of the midsole arch cavity. The raised arch shaped support piece is adapted to the natural curvature of a human foot. The anti-fatigue shoe apparatus includes a midsole arch liner that is adhered to and located within the bottom midsole surface of the midsole arch cavity. A midsole toe cavity is formed in a substantially elliptical shape and located within the bottom midsole surface of the midsole toe section. The midsole toe cavity is lined with a midsole toe liner. A midsole heel cavity is formed in a substantially oval shape and is located within the bottom midsole surface of the midsole heel section. The midsole heel cavity is also lined with a midsole heel liner. In accordance with the present invention, a toe comfort zone is located within the midsole toe cavity and a heel comfort zone is located within the midsole heel cavity. The anti-fatigue shoe apparatus includes an insole that has a top insole surface oriented opposite a bottom insole surface. The bottom insole surface is adhered to the top midsole surface and the bottom midsole surface is adhered to the top outsole surface.

(18) In one embodiment, the outsole of the anti-fatigue shoe apparatus is made out of a material that is either rubber, leather, polyurethane, or a combination of those materials.

(19) In another embodiment, the midsole of the anti-fatigue shoe apparatus is made out of a material that is either ethyl vinyl acetate, plastic, rubber, polyurethane, or a combination of those materials.

(20) In yet another embodiment, the raised arch shaped support piece of the anti-fatigue shoe apparatus is made out of a material that is either ethyl vinyl acetate, plastic, rubber, polyurethane foam, polyurethane gel, memory foam, vinyl, cork, plastic, or a combination of those materials.

(21) In one embodiment, the midsole arch liner of the anti-fatigue shoe apparatus is made out of a material that is either rubber, leather, plastic, polyurethane, or a combination of those materials.

(22) In another embodiment, the midsole toe liner and the midsole heel liner of the anti-fatigue shoe apparatus are made out of a material that is either rubber, leather, plastic, polyurethane, or a combination of those materials.

(23) In yet another embodiment, the midsole toe cavity and the midsole heel cavity are reinforced with a spray on material that consists of polyurethane, polyamide, polyacrylonitrile, or a combination of those materials.

(24) In one embodiment, the toe comfort zone and the heel comfort zone of the anti-fatigue shoe apparatus are made out of a material that is either rubber, polyurethane foam, polyurethane gel, memory foam, vinyl, cork, plastic, or a combination of those materials.

(25) In another embodiment, the anti-fatigue shoe apparatus further includes a tubular injection canal aperture that is located in the outsole. The tubular injection canal aperture leads to an outsole central injection canal. In accordance with the present embodiment, a heel comfort zone canal rises vertically from the outsole central injection canal and terminates within the midsole heel cavity through a midsole heel cavity injection aperture. A toe comfort zone canal also rises vertically from the outsole central injection canal and terminates within the midsole toe cavity through a midsole toe cavity injection aperture. In accordance with this embodiment of the anti-fatigue shoe apparatus, the outsole central injection canal, the heel comfort zone canal, and the toe comfort zone canal are adapted to receive and retain the material that the toe comfort zone and the heel comfort zone are made out of in liquid form.

(26) In another preferred embodiment of the invention, an anti-fatigue shoe apparatus is presented. The anti-fatigue shoe apparatus includes an outsole that has a top outsole surface that is oriented opposite a bottom outsole surface. The bottom outsole surface is designed to grip the ground. The anti-fatigue shoe apparatus further includes a midsole that is designed to reduce torque. The midsole has a top midsole surface that is oriented opposite a bottom midsole surface. In accordance with the present invention, a midsole arch cavity is formed centrally within the bottom midsole surface of the midsole and is located between a midsole toe section that is oriented laterally opposite a midsole heel section. A raised arch shaped support piece is located within and adhered to the bottom midsole surface of the midsole arch cavity. The raised arch shaped support piece is designed to fit the natural curvature of a human foot. The anti-fatigue shoe apparatus includes a midsole arch liner adhered to and located within the bottom midsole surface of the midsole arch cavity. A midsole toe aperture is formed in a substantially elliptical shape and oriented through the midsole toe section and a midsole heel aperture is formed in a substantially oval shape and oriented through the midsole heel section. In accordance with the present invention, a toe comfort zone is located within the midsole toe aperture and surrounded by the top midsole surface. Furthermore, a top surface of the toe comfort zone is lined with a toe comfort zone fabric that is designed for direct contact with a ball of the human foot. A heel comfort zone is located within the midsole heel aperture and is also surrounded by the top midsole surface. A top surface of the heel comfort zone is lined with a heel comfort zone fabric that is designed for direct contact with a heel of the human foot and the bottom midsole surface is adhered to the top outsole surface.

(27) In one embodiment, the anti-fatigue shoe apparatus further includes an insole having a top insole surface oriented opposite a bottom insole surface. In such embodiment, the bottom insole surface is adhered to the top midsole surface.

(28) In another embodiment, the outsole is constructed from at least one of or a combination of materials that include rubber, leather, and polyurethane.

(29) In yet another embodiment, the midsole is constructed from at least one of or a combination of materials that include ethyl vinyl acetate, plastic, rubber, and polyurethane.

(30) In one embodiment, the raised arch shaped support piece is constructed from at least one of or a combination of materials that include ethyl vinyl acetate, plastic, rubber, polyurethane foam, polyurethane gel, memory foam, vinyl, cork, and plastic.

(31) In another embodiment, the midsole arch liner is constructed from at least one of or a combination of materials that include rubber, leather, plastic, and polyurethane.

(32) In yet another embodiment, the toe comfort zone fabric and the heel comfort zone fabric are constructed from at least one of or a combination of materials that include petroleum, polyurethane, and leather.

(33) In one embodiment, the toe comfort zone and the heel comfort zone are constructed from at least one of or a combination of materials that include rubber, polyurethane foam, polyurethane gel, memory foam, vinyl, cork, and plastic.

(34) In another related embodiment, the anti-fatigue shoe apparatus further includes a tubular injection canal aperture located in the outsole. The tubular injection canal aperture leads to an

outsole central injection canal, where a heel comfort zone canal rises vertically from the outsole central injection canal and terminates within the midsole heel aperture through a midsole heel injection aperture. In accordance with such embodiment of the invention, a toe comfort zone canal rises vertically from the outsole central injection canal and terminates within the midsole toe aperture through a midsole toe injection aperture. The outsole central injection canal, the heel comfort zone canal, and the toe comfort zone canal are designed to receive and retain the material of the toe comfort zone and the heel comfort zone in liquid form.

(35) In accordance with yet another preferred embodiment of the invention, an anti-fatigue shoe apparatus is presented. The anti-fatigue shoe apparatus includes an outsole that has a top outsole surface oriented opposite a bottom outsole surface. The bottom outsole surface is designed to grip the ground and a midsole is designed to reduce torque. The midsole has a top midsole surface oriented opposite a bottom midsole surface. A midsole arch cavity is formed centrally within the bottom midsole surface of the midsole and located between a midsole toe section that is oriented laterally opposite a midsole heel section. In accordance with the present invention, the anti-fatigue shoe apparatus includes a raised arch shaped support piece located within and adhered to the bottom midsole surface of the midsole arch cavity, and the raised arch shaped support piece is designed to receive the shape of the natural curvature of the arch of a human foot. Furthermore, a midsole toe cavity is formed in a substantially elliptical shape and located within the bottom midsole surface of the midsole toe section. A midsole heel cavity is formed in a substantially oval shape and located within the bottom midsole surface of the midsole heel section. The anti-fatigue shoe apparatus includes an extruding toe comfort zone that is adhered to and located within the midsole toe cavity and an extruding heel comfort zone that is adhered to and located within the midsole heel cavity. In accordance with the present invention, the anti-fatigue shoe apparatus also includes an outsole toe cavity that is formed in a substantially elliptical shape and is located within the top outsole surface. The outsole toe cavity is designed to receive a bottom surface of the extruding toe comfort zone and is aligned with the midsole toe section. An outsole arch cavity is formed centrally within the top outsole surface of the outsole, and the outsole arch cavity is designed to receive and retain the raised arch shaped support. In accordance with the present invention, the anti-fatigue shoe apparatus also includes an outsole heel cavity that is formed in a substantially oval shape and oriented within the top outsole surface. The outsole heel cavity is designed to receive a bottom surface of the extruding heel comfort zone and is aligned with the midsole heel section. The bottom midsole surface is adhered to the top outsole surface.

(36) In one embodiment, the anti-fatigue shoe apparatus further includes an insole that has a top insole surface oriented opposite a bottom insole surface and the bottom insole surface is adhered to the top midsole surface.

(37) In another embodiment, the outsole is constructed from at least one of or a combination of materials that include rubber, leather, and polyurethane.

(38) In yet another embodiment, the midsole is constructed from at least one of or a combination of materials that include ethyl vinyl acetate, plastic, rubber, and polyurethane.

(39) In one embodiment, the raised arch shaped support piece is constructed from at least one of or a combination of materials that include ethyl vinyl acetate, plastic, rubber, polyurethane foam, polyurethane gel, memory foam, vinyl, cork, and plastic.

(40) In another embodiment, the midsole toe cavity, the midsole heel cavity, the outsole toe cavity, and the outsole heel cavity are reinforced with a spray on material which is made of at least one of or a combination of materials that include polyurethane, polyamide, and polyacrylonitrile.

(41) In yet another embodiment, the extruding toe comfort zone and the extruding heel comfort zone are constructed from at least one of or a combination of materials that include rubber, polyurethane foam, polyurethane gel, memory foam, vinyl, cork, and plastic.

(42) In another related embodiment, the anti-fatigue shoe apparatus further includes a tubular injection canal aperture located in the outsole. The tubular injection canal aperture opens to a

central injection canal. An outsole heel injection canal rises vertically from the central injection canal and terminates within the outsole heel cavity through a heel cavity injection aperture. In accordance with such embodiment of the invention, an outsole toe injection canal rises vertically from the central injection canal and terminates within the outsole toe cavity through an outsole toe cavity injection aperture. The central injection canal is designed to be substantially U-shaped and located below the outsole arch cavity. The outsole heel injection canal, the central injection canal, and the outsole toe injection canal is designed to receive and retain the material of the extruding toe comfort zone and the extruding heel comfort zone in liquid form.

(43) In accordance with an additional preferred embodiment of the present invention, an anti-fatigue shoe apparatus is presented. The anti-fatigue shoe apparatus includes an outsole that has a top outsole surface oriented opposite a bottom outsole surface. The bottom outsole surface is designed to grip the ground. The anti-fatigue shoe apparatus also includes a midsole that has a top midsole surface oriented opposite a bottom midsole surface. The top midsole surface includes a midsole toe cavity that is oriented laterally opposite from a midsole heel cavity. The midsole toe cavity is formed in a substantially elliptical shape and located within the top midsole surface, where the midsole heel cavity is formed in a substantially oval shape and is located within the top midsole surface. In accordance with the present invention, the anti-fatigue shoe apparatus includes an insole that has a top insole surface oriented opposite a bottom insole surface. An insole arch cavity is formed centrally within the bottom insole surface and the insole arch cavity is located between an insole toe section of the bottom insole surface which is oriented laterally opposite from an insole heel section of the bottom insole surface. The anti-fatigue shoe apparatus includes a raised arch shaped support piece located within and adhered to the bottom insole surface of the insole arch cavity. A midsole arch cavity is centrally formed within the top midsole surface and the midsole arch cavity is located between the midsole toe cavity and the midsole heel cavity. The midsole arch cavity is formed to retain the raised arch shaped support piece. The raised arch shaped support piece is designed to receive the shape of a natural curvature of the arch of a human foot. In accordance with the present invention, an insole toe cavity is formed in a substantially elliptical shape and located within the bottom insole surface of the insole toe section. Included is an insole heel cavity that is formed in a substantially oval shape and located within the bottom insole surface of the insole heel section. An extruding toe comfort zone is adhered to and located within the insole toe cavity and an extruding heel comfort zone is adhered to and located within the insole heel cavity. The midsole toe cavity is designed to receive a bottom surface of the extruding toe comfort zone, and the midsole heel cavity is designed to receive a bottom surface of the extruding heel comfort zone. The bottom insole surface is adhered to the bottom midsole surface and the bottom midsole surface is adhered to the top outsole surface.

(44) In one embodiment, the outsole is constructed from at least one of or a combination of materials that include rubber, leather, and polyurethane.

(45) In another embodiment, the midsole is constructed from at least one of or a combination of materials that include ethyl vinyl acetate, plastic, rubber, and polyurethane.

(46) In yet another embodiment, the insole is constructed from at least one of or a combination of materials that include ethyl vinyl acetate, plastic, rubber, and polyurethane.

(47) In one embodiment, the raised arch shaped support piece is constructed from at least one of or a combination of materials that include ethyl vinyl acetate, plastic, rubber, polyurethane foam, polyurethane gel, memory foam, vinyl, cork, and plastic.

(48) In another embodiment, the insole toe cavity, the insole heel cavity, the midsole toe cavity, and the midsole heel cavity are reinforced with a spray on material which is made of at least one of or a combination of materials that include polyurethane, polyamide, and polyacrylonitrile.

(49) In yet another embodiment, the extruding toe comfort zone and the extruding heel comfort zone are constructed from at least one of or a combination of materials that include rubber, polyurethane foam, polyurethane gel, memory foam, vinyl, cork, and plastic.

(50) In another related embodiment, the anti-fatigue shoe apparatus further includes a tubular injection canal aperture located in the outsole. The tubular injection canal aperture leads to an outsole central injection canal and a heel comfort zone canal rises vertically from the outsole central injection canal that terminates within the midsole heel cavity through a midsole heel cavity injection aperture. A toe comfort zone canal rises vertically from the outsole central injection canal and terminates within the midsole toe cavity through a midsole toe cavity injection aperture. In accordance with such embodiment, the outsole central injection canal, the heel comfort zone canal, and the toe comfort zone canal are designed to receive and retain the material of the extruding toe comfort zone and the extruding heel comfort zone in liquid form.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Illustrative embodiments of the present invention are described herein with reference to the accompanying drawings, in which:

(2) FIG. 1 is a right side elevation view of an anti-fatigue shoe apparatus within a work boot, according to a preferred embodiment of the present invention;

(3) FIG. 2 is a top, rear, left side exploded perspective view of an anti-fatigue shoe apparatus, according to a preferred embodiment of the present invention;

(4) FIG. 3 is a right side cross-sectional elevation view thereof, according to preferred embodiments;

(5) FIG. 4 is a top, rear, left side exploded perspective view of an anti-fatigue shoe apparatus, according to a preferred embodiment of the present invention;

(6) FIG. 5 is a right side elevation view of an anti-fatigue shoe apparatus within a work boot, according to a preferred embodiment of the present invention;

(7) FIG. 6 is a top, rear, left side exploded perspective view of an anti-fatigue shoe apparatus, according to a preferred embodiment of the present invention;

(8) FIG. 7 is a right side cross-sectional elevation view thereof, according to preferred embodiments;

(9) FIG. 8 is a top, rear, left side exploded perspective view thereof, with the insole option included, according to a preferred embodiment of the present invention;

(10) FIG. 9 is a right side cross-sectional elevation view thereof, according to preferred embodiments;

(11) FIG. 10 is a top, rear, left side exploded perspective view thereof, with the insole option included and the injection canals for comfort zone materials, according to a preferred embodiment of the present invention;

(12) FIG. 11 is a right side elevation view of an anti-fatigue shoe apparatus within a work boot, according to a preferred embodiment of the present invention;

(13) FIG. 12 is a top, rear, left side exploded perspective view of an anti-fatigue shoe apparatus, according to a preferred embodiment of the present invention;

(14) FIG. 13 is a right side cross-sectional elevation view thereof, according to preferred embodiments;

(15) FIG. 14 is a top, rear, left side exploded perspective view thereof, with the insole option included and the injection canals for comfort zone materials, according to a preferred embodiment of the present invention;

(16) FIG. 15 is a right side cross-sectional elevation view thereof, according to preferred embodiments;

(17) FIG. 16 is a top, rear, left side exploded perspective view thereof, with the insole option included and the injection canals for comfort zone materials, according to a preferred embodiment

of the present invention;

(18) FIG. **17** is a right side elevation view of an anti-fatigue shoe apparatus within a work boot, according to a preferred embodiment of the present invention;

(19) FIG. **18** is a top, rear, left side exploded perspective view of an anti-fatigue shoe apparatus, according to a preferred embodiment of the present invention;

(20) FIG. **19** is a right side cross-sectional elevation view thereof, according to preferred embodiments; and

(21) FIG. **20** is a top, rear, left side exploded perspective view thereof, with the injection canals for comfort zone materials, according to a preferred embodiment of the present invention.

DETAILED DESCRIPTION

(22) For a further understanding of the nature and function of the embodiments, reference should be made to the following detailed description. Detailed descriptions of the embodiments are provided herein, as well as, the best mode of carrying out and employing the present invention. It will be readily appreciated that the embodiments are well adapted to carry out and obtain the ends and features mentioned as well as those inherent herein. It is to be understood, however, that the present invention may be embodied in various forms. Therefore, persons of ordinary skill in the art will realize that the following disclosure is illustrative only and not in any way limiting, as the specific details disclosed herein provide a basis for the claims and a representative basis for teaching to employ the present invention in virtually any appropriately detailed system, structure, or manner. It should be understood that the devices, materials, methods, procedures, and techniques described herein are presently representative of various embodiments. Other embodiments of the disclosure will readily suggest themselves to such skilled persons having the benefit of this disclosure.

(23) Referring initially to FIGS. **1-4**, the basic constructional details and principles of operation of an anti-fatigue shoe apparatus **100** according to a preferred embodiment of the present invention will be discussed.

(24) FIGS. **1-4** illustrate a preferred embodiment of the anti-fatigue shoe apparatus **100** that includes an outsole **102** that has a top outsole surface **104** that is opposite a bottom outsole surface **106**. The bottom outsole surface **106** is adapted to grip the ground. Although the present invention provides for increased flexibility of work footwear, the anti-fatigue shoe apparatus **100** has a midsole **108** that is adapted to reduce torque. The midsole **108** has a top midsole surface **110** that is opposite a bottom midsole surface **112**.

(25) In accordance with the present invention, as illustrated in FIGS. **2** and **3**, a midsole arch cavity **114** is centrally formed within the bottom midsole surface **112** of the midsole **108** and is located between a midsole toe section **116** that is oriented laterally opposite of a midsole heel section **118**. A raised arch shaped support piece **120** is oriented within and adhered to the bottom midsole surface **112** of the midsole arch cavity **114**. The raised arch shaped support piece **120** is adapted to the natural curvature of a human foot. The anti-fatigue shoe apparatus **100** includes a midsole arch liner **122** that is adhered to and located within the bottom midsole surface **112** of the midsole arch cavity **114**, as shown in FIG. **2**, the midsole arch liner **122** is located between the midsole bottom surface **112** and the raised arch shaped support piece **120**. A midsole toe cavity **124** is formed in a substantially elliptical shape and located within the bottom midsole surface **112** of the midsole toe section **116**. The midsole toe cavity **124** is lined with a midsole toe liner **126**. A midsole heel cavity **128** is formed in a substantially oval shape and is located within the bottom midsole surface **112** of the midsole heel section **118**. The midsole heel cavity **128** is also lined with a midsole heel liner **130**.

(26) In accordance with the present invention, as illustrated in FIGS. **2** and **3**, a toe comfort zone **132** is located within the midsole toe cavity **124** and a heel comfort zone **134** is located within the midsole heel cavity **128**. The anti-fatigue shoe apparatus **100** includes an insole **136** that has a top insole surface **138** oriented opposite a bottom insole surface **140**. The bottom insole surface **140** is

adhered to the top midsole surface **110** and the bottom midsole surface **112** is adhered to the top outsole surface **104**.

(27) In one embodiment, the outsole **102**, shown in FIGS. **1-4**, of the anti-fatigue shoe apparatus **100** is formed from a material selected from the group consisting of rubber, leather, and polyurethane.

(28) In another embodiment, the midsole **108**, shown in FIGS. **1-4**, of the anti-fatigue shoe apparatus **100** is formed from a material selected from the group consisting of ethyl vinyl acetate, plastic, rubber, and polyurethane.

(29) In yet another embodiment, as illustrated in FIGS. **2-4**, the raised arch shaped support piece **120** of the anti-fatigue shoe apparatus **100** is formed from a material selected from the group consisting of ethyl vinyl acetate, plastic, rubber, polyurethane foam, polyurethane gel, memory foam, vinyl, cork, and plastic.

(30) In one embodiment, as shown in FIGS. **2-4**, the midsole arch liner **122** of the anti-fatigue shoe apparatus **100** is formed from a material selected from the group consisting of rubber, leather, plastic, and polyurethane.

(31) In another embodiment, as shown in FIGS. **2-4**, the midsole toe liner **126** and the midsole heel liner **130** of the anti-fatigue shoe apparatus **100** are formed from a material selected from the group consisting of rubber, leather, plastic, and polyurethane.

(32) In yet another embodiment, the midsole toe cavity **124** and the midsole heel cavity **128** are reinforced with a spray on material selected from the group consisting of polyurethane, polyamide, and polyacrylonitrile, and combinations thereof.

(33) In one embodiment, illustrated in FIGS. **2-4**, the toe comfort zone **132** and the heel comfort zone **134** of the anti-fatigue shoe apparatus **100** are formed from a material selected from the group consisting of rubber, polyurethane form, polyurethane gel, memory foam, vinyl, cork, and plastic.

(34) Turning to FIG. **4**, in another embodiment, the anti-fatigue shoe apparatus **100** further includes a tubular injection canal aperture **142** that is located in the outsole **102**. The tubular injection canal aperture **142** leads to an outsole central injection canal **144**. In accordance with the present embodiment, a heel comfort zone canal **146** rises vertically from the outsole central injection canal **144** and terminates within the midsole heel cavity **128** through a midsole heel cavity injection aperture **148**. A toe comfort zone canal **150** also rises vertically from the outsole central injection canal **144** and terminates within the midsole toe cavity **124** through a midsole toe cavity injection aperture **152**. In accordance with this embodiment of the anti-fatigue shoe apparatus **100**, the outsole central injection canal **144**, the heel comfort zone canal **146**, and the toe comfort zone canal **150** are adapted to receive and retain the material that the toe comfort zone **132** and the heel comfort zone **134** are made out of in liquid form.

(35) As illustrated in FIGS. **5-10**, preferred embodiments of an anti-fatigue shoe apparatus **200** are provided for.

(36) In such preferred embodiment of the invention, referring to FIGS. **5-7**, the anti-fatigue shoe apparatus **200** includes an outsole **202** that has a top outsole surface **204** that is oriented opposite a bottom outsole surface **206**. The bottom outsole surface **206** is designed to grip the ground. The anti-fatigue shoe apparatus **200** further includes a midsole **208** that is designed to reduce torque. As illustrated in FIGS. **6-7**, the midsole **208** has a top midsole surface **210** that is oriented opposite a bottom midsole surface **212**.

(37) In accordance with the present invention, a midsole arch cavity **214** is formed centrally within the bottom midsole surface **212** of the midsole **208** and is located between a midsole toe section **216** that is oriented laterally opposite a midsole heel section **218**, as shown in FIG. **6**. A raised arch shaped support piece **220** is located within and adhered to the bottom midsole surface **212** of the midsole arch cavity **214**. The raised arch shaped support piece **220** is designed to fit the natural curvature of a human foot.

(38) The anti-fatigue shoe apparatus **200**, as shown in FIGS. **6-9**, includes a midsole arch liner **222**

adhered to and located within the bottom midsole surface **212** of the midsole arch cavity **214**. A midsole toe aperture **224** is formed in a substantially elliptical shape and oriented through the midsole toe section **216** and a midsole heel aperture **226** is formed in a substantially oval shape and oriented through the midsole heel section **218**.

(39) In accordance with the present invention, as shown in FIGS. **6-9**, a toe comfort zone **228** is located within the midsole toe aperture **224** and surrounded by the top midsole surface **210**.

Furthermore, a top surface **236** of the toe comfort zone **228** is lined with a toe comfort zone fabric **232** that is designed for direct contact with a ball of the human foot. A heel comfort zone **234** is located within the midsole heel aperture **226** and is also surrounded by the top midsole surface **210**. A top surface **236** of the heel comfort zone **234** is lined with a heel comfort zone fabric **238** that is designed for direct contact with a heel of the human foot and the bottom midsole surface **212** is adhered to the top outsole surface **204**.

(40) In one embodiment, turning to FIGS. **8** and **9**, the anti-fatigue shoe apparatus **200** further includes an insole **240** having a top insole surface **242** oriented opposite a bottom insole surface **244**. In such embodiment, the bottom insole surface **244** is adhered to the top midsole surface **210**.

(41) In another embodiment, the outsole **202**, as shown in FIGS. **5-10**, is formed from a material selected from the group consisting of rubber, leather, and polyurethane.

(42) In yet another embodiment, as shown in FIGS. **5-10**, the midsole **208** is formed from a material selected from the group consisting of ethyl vinyl acetate, plastic, rubber, and polyurethane.

(43) In one embodiment, as illustrated in FIGS. **6-10**, the raised arch shaped support piece **220** is formed from a material selected from the group consisting of ethyl vinyl acetate, plastic, rubber, polyurethane foam, polyurethane gel, memory foam, vinyl, cork, and plastic.

(44) In another embodiment, shown in FIGS. **6-10**, the midsole arch liner **222** is formed from a material selected from the group consisting of rubber, leather, plastic, and polyurethane.

(45) In yet another embodiment, illustrated in FIGS. **6-10**, the toe comfort zone fabric **232** and the heel comfort zone fabric **238** are formed from a material selected from the group consisting of petroleum, polyurethane, and leather, and combinations thereof.

(46) In one embodiment, shown in FIGS. **6-10**, the toe comfort zone **228** and the heel comfort zone **234** are formed from a material selected from the group consisting of rubber, polyurethane form, polyurethane gel, memory foam, vinyl, cork, and plastic.

(47) Turning to FIG. **10**, in another related embodiment, the anti-fatigue shoe apparatus **200** further includes a tubular injection canal aperture **246** located in the outsole **202**. The tubular injection canal aperture **246** leads to an outsole central injection canal **248**, where a heel comfort zone canal **250** rises vertically from the outsole central injection canal **248** and terminates within the midsole heel aperture **226** through a midsole heel injection aperture **252**. In accordance with such embodiment of the invention, a toe comfort zone canal **254** rises vertically from the outsole central injection canal **248** and terminates within the midsole toe aperture **224** through a midsole toe injection aperture **256**. The outsole central injection canal **248**, the heel comfort zone canal **250**, and the toe comfort zone canal **254** are designed to receive and retain the material of the toe comfort zone **228** and the heel comfort zone **234** in liquid form.

(48) As illustrated in FIGS. **11-16**, additional preferred embodiments of an anti-fatigue shoe apparatus **300** are provided for.

(49) In such additional preferred embodiment of the invention, referring to FIGS. **11-13**, the anti-fatigue shoe apparatus **300** includes an outsole **302** that has a top outsole surface **304** oriented opposite a bottom outsole surface **306**. The bottom outsole surface **306** is designed to grip the ground and a midsole **308** is designed to reduce torque. As shown in FIGS. **12-15**, the midsole **308** has a top midsole surface **310** oriented opposite a bottom midsole surface **312**. A midsole arch cavity **314** is formed centrally within the bottom midsole surface **312** of the midsole **308** and located between a midsole toe section **316** that is oriented laterally opposite a midsole heel section **318**.

(50) In accordance with the present invention, shown in FIGS. 12-16, the anti-fatigue shoe apparatus 300 includes a raised arch shaped support piece 320 located within and adhered to the bottom midsole surface 312 of the midsole arch cavity 314, and the raised arch shaped support piece 320 is designed to receive the shape of the natural curvature of the arch of a human foot. Furthermore, a midsole toe cavity 322 is formed in a substantially elliptical shape and located within the bottom midsole surface 312 of the midsole toe section 316. A midsole heel cavity 324 is formed in a substantially oval shape and located within the bottom midsole surface 312 of the midsole heel section 318. The anti-fatigue shoe apparatus 300 includes an extruding toe comfort zone 326 that is adhered to and located within the midsole toe cavity 322 and an extruding heel comfort zone 328 that is adhered to and located within the midsole heel cavity 324.

(51) Continuing to refer to FIGS. 12-16, in accordance with the present invention, the anti-fatigue shoe apparatus 300 also includes an outsole toe cavity 330 that is formed in a substantially elliptical shape and is located within the top outsole surface 304. The outsole toe cavity 330 is designed to receive a bottom surface 338 of the extruding toe comfort zone 326 and is aligned with the midsole toe section 316. An outsole arch cavity 334 is formed centrally within the top outsole surface 304 of the outsole 302, and the outsole arch cavity 334 is designed to receive and retain the raised arch shaped support 320.

(52) In accordance with the present invention, shown in FIGS. 12-16, the anti-fatigue shoe apparatus 300 also includes an outsole heel cavity 336 that is formed in a substantially oval shape and oriented within the top outsole surface 304. The outsole heel cavity 336 is designed to receive a bottom surface 338 of the extruding heel comfort zone 328 and is aligned with the midsole heel section 318. The bottom midsole surface 312 is adhered to the top outsole surface 304.

(53) In one embodiment, reflected in FIGS. 14-16, the anti-fatigue shoe apparatus 300 further includes an insole 340 that has a top insole surface 342 oriented opposite a bottom insole surface 344 and the bottom insole surface 344 is adhered to the top midsole surface 310.

(54) In another embodiment, shown in FIGS. 11-16, the outsole 302 is formed from a material selected from the group consisting of rubber, leather, and polyurethane.

(55) In yet another embodiment, as reflected in FIGS. 11-16, the midsole 308 is formed from a material selected from the group consisting of ethyl vinyl acetate, plastic, rubber, and polyurethane.

(56) In one embodiment, illustrated in FIGS. 12-16, the raised arch shaped support piece 320 is formed from a material selected from the group consisting of ethyl vinyl acetate, plastic, rubber, polyurethane foam, polyurethane gel, memory foam, vinyl, cork, and plastic.

(57) In another embodiment, as reflected in FIGS. 12-16, the midsole toe cavity 322, the midsole heel cavity 324, the outsole toe cavity 330, and the outsole heel cavity 336 are reinforced with a spray on material selected from the group consisting of polyurethane, polyamide, and polyacrylonitrile, and combinations thereof.

(58) In yet another embodiment, the extruding toe comfort zone 326 and the extruding heel comfort zone 328, such as those in FIGS. 12-16, are formed from a material selected from the group consisting of rubber, polyurethane form, polyurethane gel, memory foam, vinyl, cork, and plastic.

(59) Turning to FIG. 16, in another related embodiment, the anti-fatigue shoe apparatus 300 further includes a tubular injection canal aperture 346 located in the outsole 302. The tubular injection canal aperture 346 opens to a central injection canal 348. An outsole heel injection canal 350 rises vertically from the central injection canal 348 and terminates within the outsole heel cavity 336 through a heel cavity injection aperture 352.

(60) In accordance with such embodiment of the invention, as illustrated in FIG. 16, an outsole toe injection canal 354 rises vertically from the central injection canal 348 and terminates within the outsole toe cavity 330 through an outsole toe cavity injection aperture 356. The central injection canal 348 is designed to be substantially U-shaped and located below the outsole arch cavity 334. The outsole heel injection canal 350, the central injection canal 348, and the outsole toe injection canal 354 is designed to receive and retain the material of the extruding toe comfort zone 326 and

the extruding heel comfort zone **328** in liquid form.

(61) As illustrated in FIGS. **17-20**, another additional preferred embodiment of an anti-fatigue shoe apparatus **400** is provided.

(62) In such additional preferred embodiment of the invention, referring to FIGS. **17-19**, the anti-fatigue shoe apparatus **400** includes an outsole **402** that has a top outsole surface **404** oriented opposite a bottom outsole surface **406**. The bottom outsole surface **406** is designed to grip the ground. The anti-fatigue shoe apparatus **400** also includes a midsole **408** that has a top midsole surface **410** oriented opposite a bottom midsole surface **412**. As illustrated in FIGS. **18-20**, the top midsole surface **410** includes a midsole toe cavity **414** that is oriented laterally opposite from a midsole heel cavity **416**. The midsole toe cavity **414** is formed in a substantially elliptical shape and located within the top midsole surface **410**, where the midsole heel cavity **416** is formed in a substantially oval shape and is located within the top midsole surface **410**.

(63) Continuing to refer to FIGS. **18-20**, in accordance with the present invention, the anti-fatigue shoe apparatus **400** includes an insole **418** that has a top insole surface **420** oriented opposite a bottom insole surface **422**. An insole arch cavity **424** is formed centrally within the bottom insole surface **422** and the insole arch cavity **424** is located between an insole toe section **426** of the bottom insole surface **422** which is oriented laterally opposite from an insole heel section **428** of the bottom insole surface **422**.

(64) The anti-fatigue shoe apparatus **400**, referring to FIGS. **18-20**, includes a raised arch shaped support piece **430** located within and adhered to the bottom insole surface **422** of the insole arch cavity **424**. A midsole arch cavity **432** is centrally formed within the top midsole surface **410** and the midsole arch cavity **432** is located between the midsole toe cavity **414** and the midsole heel cavity **416**. The midsole arch cavity **432** is formed to retain the raised arch shaped support piece **430**. The raised arch shaped support piece **430** is designed to receive the shape of a natural curvature of the arch of a human foot.

(65) Continuing to refer to FIGS. **18-20**, in accordance with the present invention, an insole toe cavity **434** is formed in a substantially elliptical shape and located within the bottom insole surface **422** of the insole toe section **426**. Included is an insole heel cavity **436** that is formed in a substantially oval shape and located within the bottom insole surface **422** of the insole heel section **428**. An extruding toe comfort zone **438** is adhered to and located within the insole toe cavity **434** and an extruding heel comfort zone **440** is adhered to and located within the insole heel cavity **436**.

(66) The midsole toe cavity **414** is designed to receive a bottom surface **442** of the extruding toe comfort zone **438**, and the midsole heel cavity **416** is designed to receive a bottom surface **444** of the extruding heel comfort zone **440**. The bottom insole surface **422** is adhered to the bottom midsole surface **412** and the bottom midsole surface **412** is adhered to the top outsole surface **404**.

(67) In one embodiment, as illustrated in FIGS. **17-20**, the outsole **402** is formed from a material selected from the group consisting of rubber, leather, and polyurethane.

(68) In another embodiment, as shown in FIGS. **17-20**, the midsole **408** is formed from a material selected from the group consisting of ethyl vinyl acetate, plastic, rubber, and polyurethane.

(69) In yet another embodiment, as reflected in FIGS. **18-20**, the insole **418** is formed from a material selected from the group consisting of ethyl vinyl acetate, plastic, rubber, and polyurethane.

(70) Continuing to refer to FIGS. **18-20**, in one embodiment, the raised arch shaped support piece **430** is formed from a material selected from the group consisting of ethyl vinyl acetate, plastic, rubber, polyurethane foam, polyurethane gel, memory foam, vinyl, cork, and plastic.

(71) In another embodiment, referring to FIGS. **18-20**, the insole toe cavity **434**, the insole heel cavity **436**, the midsole toe cavity **414**, and the midsole heel cavity **416** are reinforced with a spray on material selected from the group consisting of polyurethane, polyamide, and polyacrylonitrile, and combinations thereof.

(72) In yet another embodiment, continuing to refer to FIGS. **18-20**, the extruding toe comfort zone **438** and the extruding heel comfort zone **440** are formed from a material selected from the group

consisting of rubber, polyurethane foam, polyurethane gel, memory foam, vinyl, cork, and plastic. (73) Turning to FIG. 20, in another related embodiment, the anti-fatigue shoe apparatus 400 further includes a tubular injection canal aperture 446 located in the outsole 402. The tubular injection canal aperture 446 leads to an outsole central injection canal 448 and a heel comfort zone canal 450 rises vertically from the outsole central injection canal 448 that terminates within the midsole heel cavity 416 through a midsole heel cavity injection aperture 452. A toe comfort zone canal 454 rises vertically from the outsole central injection canal 448 and terminates within the midsole toe cavity 414 through a midsole toe cavity injection aperture 456.

(74) In accordance with such embodiment, continuing to refer to FIG. 20, the outsole central injection canal 448, the heel comfort zone canal 450, and the toe comfort zone canal 454 are designed to receive and retain the material of the extruding toe comfort zone 438 and the extruding heel comfort zone 440 in liquid form.

(75) It will be apparent to persons skilled in the relevant art that various changes in form and detail can be made without departing from the spirit and scope of the invention. While various embodiments of the present invention have been described above, it should be understood that they have been presented by way of example only, and not limitation. Thus, the breadth and scope of the present invention should not be limited by any of the above-described exemplary embodiments, but should only be defined in accordance with the following claims and their equivalents. All patents and publications discussed herein are incorporated in their entirety by reference thereto.

Claims

1. An anti-fatigue shoe apparatus, comprising: an outsole comprising a top outsole surface opposite a bottom outsole surface; wherein said bottom outsole surface adapted to grip the ground; a midsole adapted to reduce torque comprising a top midsole surface opposite a bottom midsole surface; a midsole arch cavity centrally formed within the bottom midsole surface of the midsole and disposed between a midsole toe section laterally opposite a midsole heel section; a raised arch shaped support piece disposed within and adhered to the bottom midsole surface of said midsole arch cavity, wherein said raised arch shaped support piece adapted for a natural curvature of a human foot; a midsole arch liner adhered to and disposed within the bottom midsole surface of said midsole arch cavity; a midsole toe aperture formed in a substantially elliptical shape and disposed through the midsole toe section and a midsole heel aperture formed in a substantially oval shape and disposed through the midsole heel section; a toe comfort zone disposed within the midsole toe aperture surrounded by said top midsole surface, wherein a top surface of the toe comfort zone lined with a toe comfort zone fabric adapted for direct contact with a ball of the human foot; a heel comfort zone disposed within the midsole heel aperture surrounded by said top midsole surface, wherein a top surface of the heel comfort zone lined with a heel comfort zone fabric adapted for direct contact with a heel of the human foot; and said bottom midsole surface adhered to said top outsole surface.

2. The apparatus of claim 1, further comprising an insole comprising a top insole surface opposite a bottom insole surface, said bottom insole surface adhered to the top midsole surface.

3. The apparatus of claim 1, wherein the outsole is formed from a material selected from the group consisting of rubber, leather, and polyurethane.

4. The apparatus of claim 1, wherein the midsole is formed from a material selected from the group consisting of ethyl vinyl acetate, plastic, rubber, and polyurethane.

5. The apparatus of claim 1, wherein the raised arch shaped support piece is formed from a material selected from the group consisting of ethyl vinyl acetate, plastic, rubber, polyurethane foam, polyurethane gel, memory foam, vinyl, cork, and plastic.

6. The apparatus of claim 1, wherein the midsole arch liner is formed from a material selected from the group consisting of rubber, leather, plastic, and polyurethane.

7. The apparatus of claim 1, wherein said toe comfort zone fabric and said heel comfort zone fabric are formed from a material selected from the group consisting of petroleum, polyurethane, and leather, and combinations thereof.

8. The apparatus of claim 1, wherein the toe comfort zone and the heel comfort zone are formed from a material selected from the group consisting of rubber, polyurethane foam, polyurethane gel, memory foam, vinyl, cork, and plastic.

9. The apparatus of claim 8, further comprising a tubular injection canal aperture disposed in the outsole, said tubular injection canal aperture leading to an outsole central injection canal; a heel comfort zone canal rising vertically from said outsole central injection canal terminating within the midsole heel aperture through a midsole heel injection aperture and a toe comfort zone canal rising vertically from said outsole central injection canal terminating within the midsole toe aperture through a midsole toe injection aperture, wherein said outsole central injection canal, said heel comfort zone canal, and said toe comfort zone canal adapted to receive and retain the material of the toe comfort zone and the heel comfort zone in liquid form.
